Drugs and the Kidney

- principles of drug prescribing in patients with impaired kidney function

- renal complications asso with drugs
  - examples, prevention & management

Kidneys

- 25 % of cardiac output
- GFR & renal tubular functions
- susceptibility ($\uparrow$ when there is CKD) to direct or indirect injury $\rightarrow$ AKI [$\rightarrow$ CKD]

Drug use in Patients with Impaired Renal Function

Altered clearance +/-

↑ susceptibility to adverse effects due to ↓renal fx

⇒ adjust dosage and/or frequency

Important Principles in Drug Prescription:

1. *Avoid further nephrotoxic insult*
2. *Attention to correct dose*
3. *Beware of side-effects in patients with impaired kidney function– e.g. ethambutol, acyclovir, INAH, quinolones, imipenem*

Reno-Protection in Patients with Impaired Renal Function (CKD)

1. *Prevent additional injury / insult to the kidneys*
2. *Optimal blood pressure control*
3. *Proteinuria reduction [RAAS inhibition/blockade]*
4. *SGLT2 inhibition*
5. *No smoking*
6. *Vascular risk factors*

Example – ↓dose in patients with ↓renal fx

Dipeptidyl-peptidase-IV (DPP-4) inhibitors – ↑endogenous incretin → ↑insulin & ↓glucagon in a glucose-dependent manner → blood glucose control

Linagliptin (Tradjenta) – 5 mg/D; hepatobiliary elimination; no need to change acc to renal fx

Alogliptin (Nesina) - 25 mg/D when CrCl ≥60 mL/min, 12.5 mg/D when CrCl ≥30 to <60 mL/min; 6.25 mg/D when CrCl <30 mL/min

Saxagliptin (Onglyza) – 2.5 mg [co-admin cytochrome P450 3A4/5 inhibitor] or 5 mg/D when CrCl >50 mL/min, 2.5 mg/D when CrCl ≤50 mL/min or HD (post-HD); no data in PD

Sitagliptin (Januvia) - 100 mg/D when CrCl ≥50 mL/min, 50 mg/D when CrCl ≥30 to <50 mL/min, 25 mg/D when CrCl <30 mL/min, HD (anytime) or PD

Example – ↓dose in patients with ↓renal fx

Tenofovir – HBV, HIV treatment; renal & bone side-effects; pharmaokinetic interactions with other HIV drugs

Tenofovir disoproxil fumarate (TDF)
CrCl ≥50 mL/min – 300 mg daily
CrCl 30-49 mL/min - 300 mg PO qODay
CrCl 10-29 mL/min - 300 mg PO q72-96hr
CrCl <10 mL/min - not studied
Hemodialysis - 300 mg PO q7Days

Tenofovir alafenamide (TAF) – pro-drug, higher intracellular conc and lower plasma conc vs TDF; better renal (bone) safety profile; CrCl ≥15 mL/min – 25 mg daily

Drug-induced Renal Impairment

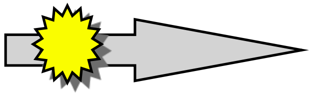
Functional

NSAID, ACEI

Structural → renal parenchymal disease

tubulointerstitial vs glomerular

NSAID

cyclo-oxygenase  PG

[vasodilation]

** hypoperfusion states - e.g. cardiac insufficiency, hypovolemia, massive haemorrhage, nephrotic, cirrhosis

RAAS Blockage & GFR

angiotensin II → vasoconstriction of efferent arteriole

renal artery stenosis, hypovolemia, cirrhosis

RAAS Blockade

(ACEI / ARB / Aldosterone Antagonist / Renin Inhibitor)

Cautions -

- renal artery stenosis [patients with vascular risk factors]
- hyperkalaemia
- metabolic acidosis

Drug-induced Nephropathy

Mechanisms

- haemodynamic (functional)
- Glomerular / Tubulointersititial (Immunological injury)
- direct toxicity (tubular cells)

Examples

Drug-induced Glomerular Diseases

e.g. ‘secondary’ membranous nephropathy

proteinuria, nephrotic syndrome

[gold salts, D-penicillamine]

e.g. minimal change disease

[NSAID]

Drug-induced Tubulo-Interstitial Nephritis

- many drugs implicated, e.g.
  antibiotics (e.g. methicillin, rifampicin)
  NSAID, immunotherapy
- may have other ‘allergic’ features (e.g. allopurinol)
- not uncommon

Drug-induced Tubulo-Interstitial Nephritis

- acute kidney injury
- may lead to permanent damage
- Management - STOP the incriminated drug
- ? Immunosuppression (usually not necessary, except in drug-induced vasculitis, immune CPI associated AKI?)

AKI due to Immune Check-Point Inhibitors

Immune checkpoint inhibitors (CPIs) – monoclonal Ab targeting molecules on the surface of T cells for the treatment of solid organ or haematologic malignancies

report of13 patients (ipilimumab 10, pembrolizumab 2, nivolumab 1)
– onset 21-245 D, one case 2 months *after* last CPI dose; uWBC 8/13; uP/Cr 0.12-0.98 g/g; extra-renal 7/13; peak Cr 3.6-7.3 mg/dL; 4 patients required *HD*; acute TIN 12/13, granulomatous 3/13, TMA 1/13; 10 treated with *steroid* – 2 complete & 7 partial *improvement*, 2 with acute TIN not given steroid did not improve

Cortazar FB, et al. Kidney Int 2016; 90: 638-47

Direct Toxicity to Renal Tubular Cells

antibiotics - aminoglycosides
some cephalosporins; quinolones

lithium, cisplatin, methotrexate, nitrosoureas

adefovir, tenofovir disoproxil fumarate (vs alafenamide)

contrast media

NSAID

Aminoglycosides

- concentrated in kidneys
- phospholipid brush border of prox tubular epith
- endocytosis; inhibit phospholipases, accumulation of lipids, modify enzyme activities, Na-K ATPase,
- reduce lysosomal membrane permeability and mitochondrial respiration

Aminoglycosides

nephrotoxicity, ototoxicity

ARF (non-oliguric initially)

preceded by polyuria, tubular dysfunction

Pathology - rarification then disappearance of brush border; enlarged lysosomes with myeloid bodies; mitochondrial swelling; tubular necrosis;

regeneration

Aminoglycosides

Risk factors for nephrotoxicity:

[note narrow therapeutic window]

- dose & duration of therapy
- Na depletion, hypovolemia, K depletion
- other nephrotoxic agents
- pre-existing renal impairment
- clinical state e.g. septicaemia

Lithium

- Nephrogenic diabetes insipidus
- CKD – chronic tubulointerstitial nephropathy (out of proportion to glomerulosclerosis, cysts and tubular dilatation)
- Acute lithium intoxication

[lithium could also lead to hypothyroidism and hyperparathyroidism]

Quiz 1:

What is nephrogenic diabetes insipidus and how does it manifest?

Cisplatin

nephrotoxicity, ototoxicity, GI, myelosuppression

dose > 100 mg/m^2 ⇨ ↑ risk of nephrotoxicity

↑ renal vasc resistance → ↓ renal plasma flow

tubular dysfunction may not correlate with ↓ CrCl

Cisplatin

Hypo Mg - 70-80 % may persist for months

Hypo Ca Hypo K

focal ATN, distal & collecting tub

dilatation of convoluted tub, casts

can be *irreversible*

Prevention - hydration 200 ml/h during and for 6 h

after cisplatin

NSAID

- sodium & fluid retention
- reduce renal blood flow and GFR
- hyperkalemia
- tubulointersitial nephritis
- minimal change nephrotic syndrome
- papillary necrosis in diabetic patients

NSAID-induced Nephrotic Syndr + AKI

minimal change glomerulopathy

acute tubulointerstitial nephritis

T lymphocytes

eosinophils (40 %)

? higher risk in elderly

One drug can lead to kidney complications through different / multiple mechanisms

Cyclosporin / Tacrolimus

Calcineurin inhibitors

modulatory effect on podocytes

Acute nephrotoxicity –

reduced intra-renal blood flow

enhanced by mTOR inhibitors

Chronic nephrotoxicity –

vasculopathy and fibrosis (TGF-beta)

blood level monitoring

Cyclosporin / Tacrolimus

Metabolic effects –

↑ K, ↑ Cl, ↓ bicarbonate

Aristocholic Acid Nephropathy

CKD

uro-epithelial malignancies

Biphosphonates

- Inhibit osteoclast-mediated bone resorption e.g. metastatic bone dis, osteoporosis
- Bound to bone (40-60%), remainder excreted unchanged through kidney (both glom filtration and active tubular secretion)

- **Tubular toxicity with ↓renal fx** may occur with iv high-dose (risk varies between different drugs)
- **Collapsing FSGS** associated with iv high-dose

Radiological Contrast

toxicity to renal tubular cells → ↓ renal fx / urine

Susceptibility / Risk Factors

- hypovolemia, renal insuff, DM, dose, myeloma, other nephrotoxic agents, elderly

Prevention – patient selection, hydration, ?N-acetyl cysteine, ?sodium bicarbonate, choose the imaging modality wisely

Nephrogenic Systemic Fibrosis [Nephrogenic Fibrosing Dermopathy]

Expansion and fibrosis of dermis → thickened and hardened skin and internal organs

MRI with gadolinium in patients with mod/severe renal failure (esp GFR <30 ml/min)

?low risk with group 2 gadolinium

Bowel preparation for colonoscopy

patients with renal impairment →

oral phospho-soda laxative may induce

hyperphosphataemia and severe hypocalcaemia

‘acute phosphate nephropathy’

Vukasin P, et al. Dis Colon Rectum 1997; 40: 497-9

Acute Phosphate Nephropathy

phosphate load (oral sodium phosphate bowel purgatives*) + dehydration

$\Rightarrow$ acute / chronic renal failure

Intra-tubular calcium phosphate precipitation

**Avoid* in high-risk patients – use alternatives
e.g. polyethylene glycol

Miscellaneous

cyclophosphamide → ↓ urinary dilution

hydralazine → drug-induced lupus

propylthiouracil → drug-induced vasculitis

metformin in CKD → lactic acidosis

Drug-induced Renal Vasculitis

- penicillin, sulphonamides, anti-thyroid drugs
- renal + extra-renal manifestations
- fibrinoid necrosis of intima and media of small/medium arteries
- periarteriolar cellular infiltrate
- Rx: Immunosuppression

Quiz 2:

Which of the following are potential side-effects of calcineurin inhibitors?

A.	hypertension	yes / no
B.	hyperglycaemia	yes / no
C.	hyperkalaemia	yes / no
D.	metabolic acidosis	yes / no
E.	nephrotoxicity	yes / no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Questions ?

TELL US WHAT

SC Whole Class Session

YOU THINK

THANK YOU!

SC
WCS

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)